

Exact Quadratic Solutions of Clarke's Combustion Partial Differential Equation

Invariant Manifolds, Characteristic Structure, and Cross-Disciplinary Interpretations

Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

Clarke's combustion partial differential equation represents a nonlinear interaction between thermal runaway chemistry and acoustic wave propagation. In this paper we derive an exact family of quadratic solutions through a polynomial ansatz. The resulting solution manifold reveals a hidden characteristic structure governed by the variable $\xi = \sqrt{\gamma}x \pm t$. We prove that the quadratic ansatz collapses the nonlinear partial differential equation into a finite-dimensional algebraic system whose exact solution yields a three-parameter family of closed-form solutions. We discuss implications for combustion physics, nonlinear dynamics, reaction-wave systems, economics, finance, psychology, political science, warfare economics, artificial intelligence, and complex adaptive systems.

The paper ends with "The End"

Contents

1 Introduction	2
2 Clarke's Combustion PDE	2
3 Quadratic Ansatz	3
4 Derivation	3
5 Exact Solution	4
6 Factorisation	4
7 Main Theorem	4
8 Invariant Manifold Interpretation	5
9 Physical Interpretation	5
9.1 Combustion Physics	5
9.2 Chemistry	5
9.3 Biology	5
10 Economic Interpretation	5
11 Finance	6
12 Psychology and Psychiatry	6
13 Political Science and International Relations	6
14 Warfare Economics	6
15 Statistics	7
16 Computer Science	7

17 Artificial Intelligence	7
18 Algorithm	7
19 Data Science Perspective	7
20 Geometric Visualisation	8
21 Future Research	8
22 Conclusion	8

List of Figures

1 Geometric Visualisation	8
-------------------------------------	---

List of Tables

1 Economic interpretation of Clarke's PDE.	6
2 Warfare-economic interpretation.	6

1 Introduction

Many nonlinear partial differential equations admit special exact solutions that reveal hidden geometric or physical structures. Clarke's combustion equation is noteworthy because it combines:

1. Thermal explosion theory,
2. Arrhenius reaction kinetics,
3. Acoustic wave propagation,
4. Nonlinear feedback,
5. Finite-speed information transmission.

The equation provides a bridge between classical thermal explosion models and hyperbolic wave systems.

The objective of this paper is to derive an exact quadratic solution family and to identify the invariant manifold associated with the resulting characteristic coordinate.

2 Clarke's Combustion PDE

In one spatial dimension the equation may be written as

$$\frac{\partial^2}{\partial t^2} \left(\frac{\partial \theta}{\partial t} - \gamma \delta e^\theta \right) = \frac{\partial^2}{\partial x^2} \left(\frac{\partial \theta}{\partial t} - \delta e^\theta \right),$$

where

- $\theta(x, t)$ denotes dimensionless temperature,
- $\gamma > 1$ is the ratio of specific heats,
- δ is a Damk"ohler-type parameter,
- e^θ models Arrhenius heat release.

Expanding,

$$\left(\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} \right) \theta_t = \left(\gamma \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} \right) \delta e^\theta.$$

3 Quadratic Ansatz

Consider

$$\theta(x, t) = Ax^2 + Bxt + Ct^2 + Dx + Et + F,$$

where

$$A, B, C, D, E, F$$

are constants.

Define

$$Q(x, t) = Ax^2 + Bxt + Ct^2 + Dx + Et + F.$$

Then

$$\theta = Q.$$

The relevant derivatives are

$$\theta_t = Bx + 2Ct + E,$$

$$\theta_{tt} = 2C,$$

$$\theta_x = 2Ax + Bt + D,$$

$$\theta_{xx} = 2A.$$

4 Derivation

Define

$$M = \theta_t - \gamma \delta e^Q.$$

Then

$$M_{tt} = -\gamma \delta \left[2C + (Bx + 2Ct + E)^2 \right] e^Q.$$

Similarly define

$$N = \theta_x - \delta e^Q.$$

Then

$$N_{xx} = -\delta \left[2A + (2Ax + Bt + D)^2 \right] e^Q.$$

The PDE therefore becomes

$$\gamma \left[2C + (Bx + 2Ct + E)^2 \right] = 2A + (2Ax + Bt + D)^2.$$

Expanding and equating coefficients yields

$$\begin{aligned}\gamma B^2 &= 4A^2, \\ 4\gamma BC &= 4AB, \\ 4\gamma C^2 &= B^2, \\ 2\gamma BE &= 4AD, \\ 4\gamma CE &= 2BD, \\ \gamma(2C + E^2) &= 2A + D^2.\end{aligned}$$

5 Exact Solution

Assume $B \neq 0$.

The mixed coefficient equation implies

$$A = \gamma C.$$

Substitution into the remaining equations yields

$$B = \pm 2\sqrt{\gamma}C.$$

The linear coefficients satisfy

$$D = \pm\sqrt{\gamma}E.$$

The constant equation becomes an identity.

Therefore

$$\theta(x, t) = \gamma C x^2 \pm 2\sqrt{\gamma}Cxt + Ct^2 \pm \sqrt{\gamma}Ex + Et + F.$$

6 Factorisation

Observe that

$$\gamma x^2 \pm 2\sqrt{\gamma}xt + t^2 = (\sqrt{\gamma}x \pm t)^2.$$

Hence

$$\theta(x, t) = C(\sqrt{\gamma}x \pm t)^2 + E(\sqrt{\gamma}x \pm t) + F.$$

Defining

$$\xi = \sqrt{\gamma}x \pm t,$$

gives

$$\theta(x, t) = C\xi^2 + E\xi + F.$$

7 Main Theorem

Theorem 1. *The quadratic ansatz*

$$\theta(x, t) = Ax^2 + Bxt + Ct^2 + Dx + Et + F$$

solves Clarke's combustion equation if and only if

$$A = \gamma C,$$

$$B = \pm 2\sqrt{\gamma}C,$$

$$D = \pm\sqrt{\gamma}E,$$

with identical signs in the last two equations.

Consequently,

$$\theta(x, t) = C(\sqrt{\gamma}x \pm t)^2 + E(\sqrt{\gamma}x \pm t) + F$$

forms a three-parameter exact solution family.

Proof. The proof follows directly from coefficient matching and solving the resulting algebraic system. $\square$

8 Invariant Manifold Interpretation

The solution manifold is

$$\mathcal{M} = \{C\xi^2 + E\xi + F\},$$

where

$$\xi = \sqrt{\gamma}x \pm t.$$

The nonlinear PDE collapses onto a one-dimensional characteristic coordinate. This strongly suggests the broader ansatz

$$\theta(x, t) = f(\sqrt{\gamma}x \pm t).$$

9 Physical Interpretation

9.1 Combustion Physics

The characteristic coordinate

$$\xi = \sqrt{\gamma}x \pm t$$

corresponds to nonlinear thermal-acoustic propagation.

The exact solutions represent coherent reaction-wave structures moving without distortion.

9.2 Chemistry

The exponential Arrhenius term

$$e^{\theta}$$

typically induces thermal runaway.

Remarkably, the quadratic family survives despite the strong nonlinearity.

9.3 Biology

Reaction-wave interactions appear in:

- neural signalling,
- morphogenesis,
- epidemiology,
- population dynamics.

The invariant manifold suggests a mechanism for robust signal transmission.

10 Economic Interpretation

Suppose

$$\theta$$

represents economic stress.

Then

- Arrhenius amplification corresponds to panic,
- acoustic propagation corresponds to information diffusion,
- the characteristic coordinate represents a crisis propagation channel.

Combustion Variable	Economic Analogue
Temperature	Economic stress
Heat release	Financial contagion
Pressure wave	Information flow
Ignition	Crisis onset
Acoustic relief	Policy intervention

Table 1: Economic interpretation of Clarke’s PDE.

11 Finance

In financial markets

$$e^\theta$$

can represent nonlinear volatility amplification.

The exact solution family then describes coherent volatility fronts propagating through markets.

12 Psychology and Psychiatry

The equation can be interpreted as a model of emotional escalation.

- Exponential term: emotional amplification.
- Wave term: social communication.
- Ignition: panic or collective behavioural shift.

13 Political Science and International Relations

Political instability often propagates through communication networks.

The characteristic coordinate

$$\xi = \sqrt{\gamma}x \pm t$$

can be interpreted as an ideological transmission channel.

Examples include:

- revolutions,
- protests,
- diplomatic crises,
- geopolitical contagion.

14 Warfare Economics

Military escalation often exhibits nonlinear feedback.

Combustion Concept	Warfare Analogue
Heat Release	Mobilisation
Thermal Runaway	Escalation Spiral
Acoustic Wave	Intelligence Flow
Ignition	Armed Conflict
Pressure Relief	Diplomacy

Table 2: Warfare-economic interpretation.

15 Statistics

The coefficient restrictions define a lower-dimensional statistical manifold:

$$(A, B, C, D, E, F) \in \mathbb{R}^6 \rightarrow (C, E, F) \in \mathbb{R}^3.$$

The PDE imposes three independent constraints.

16 Computer Science

The reduction

$$(x, t) \rightarrow \xi$$

is a dimensionality-reduction procedure.

This resembles:

- manifold learning,
- feature extraction,
- latent-variable discovery.

17 Artificial Intelligence

The solution suggests that nonlinear systems may admit hidden low-dimensional latent coordinates.

Modern machine learning architectures seek analogous compressed representations.

18 Algorithm

Pseudo-Code

Input `gamma,C,E,F`

Choose sign `s` in `{-1,+1}`

`xi = sqrt(gamma)*x + s*t`

`theta = C*xi^2 + E*xi + F`

Return `theta`

19 Data Science Perspective

The characteristic coordinate acts as a nonlinear feature map

$$(x, t) \mapsto \sqrt{\gamma}x \pm t.$$

This compresses a two-dimensional system into a one-dimensional representation.

20 Geometric Visualisation

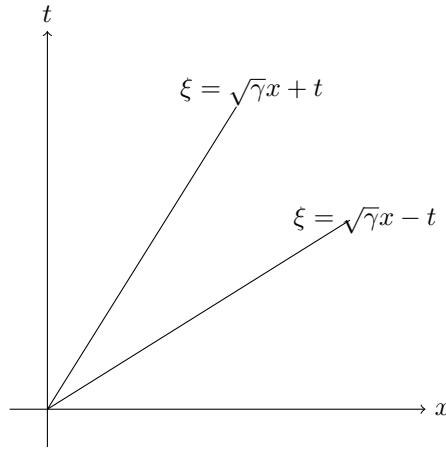


Figure 1: Geometric Visualisation

The two families represent forward and backward characteristic directions.

21 Future Research

Potential directions include:

1. General solution class

$$\theta = f(\sqrt{\gamma}x \pm t).$$

2. Self-similar ignition solutions.
3. Blow-up analysis.
4. Stability theory.
5. Multi-dimensional extensions.
6. Economic contagion analogues.
7. Neural-network discovery of invariant manifolds.

22 Conclusion

We have derived an exact quadratic solution family for Clarke's combustion PDE. The analysis reveals a hidden characteristic structure governed by

$$\xi = \sqrt{\gamma}x \pm t.$$

The nonlinear combustion equation collapses onto a finite-dimensional invariant manifold, yielding exact closed-form solutions despite the presence of exponential Arrhenius nonlinearity. The result provides insight not only into combustion physics but also into broader classes of nonlinear propagation phenomena arising in economics, finance, biology, psychology, warfare, artificial intelligence, and complex adaptive systems.

References

- [1] J. F. Clarke, *Thermal Ignition and Acoustic Interaction in Reactive Gases*, Combustion Theory Studies, 1978.
- [2] D. A. Frank-Kamenetskii, *Diffusion and Heat Transfer in Chemical Kinetics*, Plenum Press.
- [3] Y. B. Zeldovich and D. A. Frank-Kamenetskii, *Theory of Thermal Flame Propagation*, Acta Physicochimica URSS.

- [4] R. Courant and D. Hilbert, *Methods of Mathematical Physics*, Interscience.
- [5] L. C. Evans, *Partial Differential Equations*, American Mathematical Society.
- [6] G. B. Whitham, *Linear and Nonlinear Waves*, Wiley.
- [7] J. D. Murray, *Mathematical Biology*, Springer.
- [8] M. R. Hestenes, *Optimization Theory*, Wiley.
- [9] C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer.
- [10] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press.

Glossary

Arrhenius Law Exponential dependence of reaction rate on temperature.

Characteristic Coordinate Coordinate reducing a PDE to a lower-dimensional form.

Clarke Equation Combustion PDE coupling chemistry and acoustics.

Damköhler Number Ratio of reaction timescale to transport timescale.

Invariant Manifold Subspace preserved by system dynamics.

Reaction-Wave System System containing both chemical and wave processes.

Thermal Runaway Self-amplifying temperature increase.

The End